

ReTAG: A Retrieved Cellular Topologies-Augmented Graph Learning Framework

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Abstract—Retrieval-augmented generation (RAG) methods for graph learning enhance the generalization of Graph Neural Networks (GNNs) by retrieving and integrating structurally relevant subgraphs, addressing their limitations on unseen or distribution-shifted graphs. From the perspective of topology theory, cycles emerge naturally as closed arrangements of nodes and edges and serve as the boundaries of potential higher-dimensional cells. These cycles encode relational patterns and dependencies that span multiple elements, linking fine-grained (node/edge) interactions to coarser structural motifs. Neglecting such structures limits the ability of RAG methods to retrieve meaningful subgraphs and reason over the full topology of a graph. In this paper, we propose a novel Retrieved Cellular Topologies-Augmented Graph Learning Framework (ReTAG), that leverages cellular complexes to model and retrieve multi-dimensional topology-aware subgraphs, termed cellular topologies. These structures encode multi-dimensional topological interactions across nodes, edges, and higher-dimensional cells. During inference, ReTAG retrieves cellular topologies based on their topological and semantic alignment with the input graph, and integrates them via a multi-dimensional topological message-passing mechanism that enables effective propagation of topological information across dimensions. Experiments on node classification, link prediction, and graph classification show ReTAG outperforms existing methods.

Index Terms—Graph Neural Networks, Retrieval-Augmented Generation, Cell Complexes, Graph Prompt Tuning.

I. INTRODUCTION

GRAPH representation learning encodes graph-structured data into low-dimensional embeddings that capture topological and semantic information, underpinning relational modeling in domains such as social networks [1]–[3], biochemistry [4], [5], and traffic systems [6], [7]. Within this context, Graph Neural Networks (GNNs) [8], [9], leveraging message-passing architectures, have become the dominant approach, enabling more expressive relational modeling than traditional node embedding methods [10]. However, despite their effectiveness, GNNs often exhibit limited generalization to unseen graphs with substantially different topologies [11], a sensitivity that poses significant challenges for real-world applications where graph structures vary widely.

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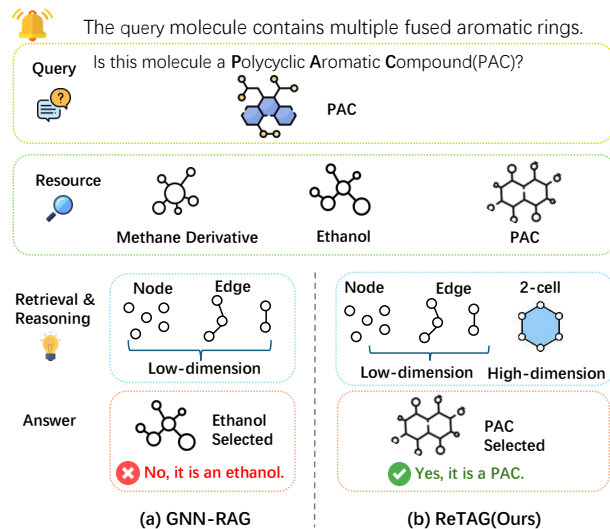


Fig. 1. RAG-based graph learning with multi-dimensional topologies.

To overcome this limitation, retrieval-augmented methods for graph learning have recently emerged as a promising approach [12], [13], drawing inspiration from retrieval-augmented generation (RAG) techniques in NLP. RAG methods enhance GNNs by retrieving external subgraphs that are structurally or semantically relevant to the input, and integrating them into the learning pipeline to provide additional contextual information. Notably, RAGRAP [13] have demonstrated enhanced adaptability by incorporating retrieved graph fragments through message-passing prompt mechanisms. However, existing RAG methods operate primarily on nodes and edges, overlooking cycles that reflect latent higher-dimensional structures critical for capturing relational patterns and complex subgraph dependencies.

In many real-world graphs (as illustrated in Fig.1), critical structural information arises not only from 0-dimensional nodes and 1-dimensional edges, but also from cyclic motifs, which form the boundaries of potential higher-dimensional cells. These cycles capture recurring relational patterns and domain-specific structural dependencies, providing inductive biases crucial for accurate interpretation and retrieval. However, conventional GNN-RAG (Fig.1(a)) methods struggle to retrieve meaningful analogues when they fail to recognize or reason over such high-dimensional structures. In contrast, retrieving cyclic motifs from external graph corpora (Fig.1(b)) can provide structurally analogous examples that support inference in novel molecules, especially when query graphs exhibit global patterns that differ significantly from those during train-

ing. Moreover, high-dimensional reasoning captures complex dependencies beyond traditional pairwise message passing.

Despite the promise of topology-aware graph learning, leveraging multi-dimensional structures for retrieval-enhanced inference faces several key challenges. First, it is non-trivial to model graph substructures across multiple dimensions while preserving their hierarchical organization. Cell complexes [14], a fundamental construct in the theory of algebraic topology [15], provide a principled framework by organizing nodes, edges, and cycles into coherent units, where cycles serve as boundaries for higher-dimensional cells, capturing richer structural motifs than the original graph. Constructing such complexes consistently across heterogeneous graphs and organizing their multi-level components into a structured knowledge base is challenging. Second, effective reasoning requires propagating information across these multi-dimensional motifs. Relational dependencies often span nodes, edges, and higher-dimensional cells, demanding a mechanism that integrates messages across dimensions while preserving topological coherence—capabilities beyond conventional GNNs or standard retrieval approaches. Third, even with suitable structures retrieved, models must maintain their internal consistency; without explicit guidance, multi-dimensional information can collapse into pairwise interactions. A dedicated objective is therefore necessary to enforce coherent representations across cells and dimensions.

In this work, we propose a novel **Retrieved Cellular Topologies-Augmented Graph Learning Framework (ReTAG)** that leverages cellular complexes to model and retrieve multi-dimensional topology-aware subgraphs, termed cellular topologies. Specifically, ReTAG first lifts input graphs into cellular complexes to capture high-dimensional topological structures, enabling the extraction of multi-dimensional topology-aware substructures. These substructures are organized into a knowledge base that encodes rich topological and semantic information beyond traditional nodes and edges. During inference, ReTAG retrieves cellular topologies that align both topologically and semantically with the input graph, and integrates them through a multi-dimensional topological message-passing mechanism designed to propagate topological information effectively across different dimensions. Furthermore, a cellular topological contrastive learning module is introduced to enhance structural discrimination by enforcing feature consistency within each 2-cell, thereby capturing topological semantics beyond conventional pairwise interactions. Extensive experiments demonstrate that ReTAG significantly outperforms existing methods across diverse graph scenarios.

Our contributions in this work are summarized as follows:

- We highlight the critical role of higher-dimensional topological structures in graph learning.
- We propose ReTAG that constructs a knowledge base of topology-aware substructures from lifted cellular complexes, and performs retrieval-augmented inference through multi-dimensional message passing to effectively integrate external topological knowledge.
- Extensive experiments across multiple graph tasks demonstrate that ReTAG significantly outperforms state-of-the-art baselines.

II. RELATED WORKS

A. Graph Representation Learning

Graph representation learning aims to capture structural and semantic information from graph-structured data. GNNs have become a dominant tool in this area by aggregating information through message passing [14], [16]–[18]. Despite their success, GNNs struggle with dynamic or multi-task settings due to rigid architectures and fixed message-passing schemes that lack task-aware modulation and flexible inference control.

To improve model flexibility and task adaptivity, recent studies have proposed graph prompt learning, inspired by prompt-based tuning in natural language processing [19], [20]. This paradigm typically involves pre-training GNNs on large-scale graph [11], [21]–[24] and guiding downstream task execution through prompt-like structures that inject task-relevant information. For example, VNT [25] introduces virtual nodes as soft prompts to encode task-level context and guide message propagation. GraphPrompt [26] proposes a task-specific readout mechanism that learns to aggregate node representations differently depending on the downstream objective. GraphPro [27] further designs spatial- and temporal-aware gating mechanisms, adapting the prompt to dynamic graph settings like recommender systems. PRODIGY [28] constructs a task graph as an explicit prompt structure and learns cross-graph in-context learning capabilities by interacting it with a data graph via a transformer-based module. ProNoG [29] proposes a novel pre-training and prompt learning framework tailored for non-homophilic graphs, introducing a conditional network that captures node-specific relational patterns to enhance downstream task performance. Graph prompt learning typically presumes structural alignment with pretraining data, hindering generalization to graphs with divergent topologies.

To overcome this limitation, recent work has drawn inspiration from RAG in NLP [12], [30], where external information is retrieved and injected into a pre-trained model to support better generalization. In the context of graph learning, a representative effort is RAGRAP [13], which enhances GNNs through a retrieval-based framework that leverages a toy-graph library to incorporate structurally or semantically similar graphs via message-passing prompting, enabling adaptation without retraining. Despite their effectiveness, existing graph RAG methods primarily rely on low-dimensional structures, overlooking higher-dimensional topologies crucial for capturing key substructures and complex relations.

B. Topological Deep Learning

Topological deep learning (TDL) builds upon early advances in Topological Signal Processing (TSP) [31]–[34], which highlighted the importance of modeling higher-order relations beyond pairwise connections. This line of work has motivated the generalization of classical graph-theoretic tools to richer topological domains, such as simplicial complexes (SCs) and cell complexes (CWs). In particular, extensions of the Weisfeiler–Lehman test to SCs and CWs [14], [18] have laid the theoretical foundations for higher-order message passing. Subsequently, a variety of neural architectures have

been proposed, including convolutional designs for SCs and CWs [35]–[40] and attentional formulations [41], [42]. To unify these efforts, the combinatorial complex (CC) framework [43] was introduced, providing a general message-passing paradigm that encompasses SCs, CWs, and hypergraphs. Parallel to these developments, Sheaf Neural Networks (SNNs) [44]–[49] have demonstrated the effectiveness of sheaf-based modeling in handling heterophily and capturing localized topological constraints.

While these works substantially extend graph representation learning to higher-dimensional domains, they largely remain confined to direct message passing within predefined complexes. Such approaches often exhibit limited generalization to unseen graphs with substantially different topologies, a limitation that poses significant challenges for real-world applications where structural variability is the norm.

III. PRELIMINARIES

a) Definition 1. (Cell Complex [44]): A **regular cell complex** is a topological space X decomposed into a collection of disjoint subspaces $\{x_\alpha\}_{\alpha \in P_X}$, referred to as *cells*, satisfying the following conditions: left=1.2em, label=1.

- 1) For each point $p \in X$, there exists an open neighborhood intersecting only finitely many cells.
- 2) For any pair of cells x_α, x_τ , the intersection $x_\tau \cap \overline{x_\alpha}$ is nonempty if and only if $x_\tau \subseteq \overline{x_\alpha}$, where $\overline{x_\alpha}$ denotes the topological closure of x_α .
- 3) Each cell x_α is homeomorphic to an open ball in $\mathbb{R}^n$ for some non-negative integer n .
- 4) (*Regularity*) The closure $\overline{x_\alpha}$ of every cell is homeomorphic to a closed ball in $\mathbb{R}^{n_\alpha}$, with the interior mapped homeomorphically onto x_α itself.

b) Definition 2: A **cellular lifting map** is a function $f : \mathcal{G} \rightarrow X$ from the space of graphs $\mathcal{G}$ to the space of regular cell complexes X , satisfying that two graphs $G_1, G_2 \in \mathcal{G}$ are isomorphic if and only if their corresponding cell complexes $f(G_1)$ and $f(G_2)$ are isomorphic. Intuitively, a cell complex is built hierarchically by first considering 0-cells (vertices), then attaching 1-cells (edges) via their endpoints, and further incorporating higher-dimensional cells—such as 2-cells—by gluing disks along cycles.

c) Definition 3. (Retrieved Cellular Topologies-Augmented Graph Learning): Given a graph $\mathcal{G}$, a cellular lifting map $f : \mathcal{G} \rightarrow X$ is conducted from the space of graphs to the space of regular cell complexes. To construct a cell complex knowledge base $X = \{X_\sigma\}$, we extract localized subcomplexes from the evolving complex X centered around a master 0-cell x_m^0 , defined as the 0-cell with the highest degree in the complex. Each subcomplex includes the set $k\text{-hop}_{X^{(0)} \cup X^{(1)}}(x_m^0)$ of 0- and 1-cells reachable within k hops from x_m^0 , as well as all 2-cells $x^2 \in X^{(2)}$ such that there exists a cell $x^1 \in k\text{-hop}_{X^{(0)} \cup X^{(1)}}(x_m^0)$ with $x^1 \prec x^2$, meaning x^1 is a face of x^2 . To enable retrieval, each cell complex X_σ is indexed by a composite **key** that includes: the timestamp τ_b , the master 0-cell embedding h_m^0 , and a two-dimensional topological characteristic h_σ^2 obtained by pooling over all 2-cells within X_σ . Retrieval is performed

via similarity over keys to obtain **values** such as task-specific output vectors $\{o_i \in \mathbb{R} \mid x_i \in X_\sigma\}$ and cell complex embeddings $\{h_i \in \mathbb{R} \mid x_i \in X_\sigma\}$.

Given a graph $\mathcal{G}$, we first split it into training and testing subsets, $\mathcal{G} = \mathcal{G}_{\text{train}} \cup \mathcal{G}_{\text{test}}$, and lift them into a regular cell complex $X = X_{\text{train}} \cup X_{\text{test}}$ via a map $f : \mathcal{G} \rightarrow X$. Each unit (e.g., a 0-cell x_i^0 , 1-cell x_{ij}^1 , or complex X_σ) has label y_i if and only if it resides in X_{train} . The goal is to predict Y_{test} by retrieving relevant cell complexes and propagating structure-aware representations.

IV. THE RETAG FRAMEWORK

In this section, we present the architecture of ReTAG (illustrated in Fig. 2), a retrieval-augmented graph learning framework built upon multi-dimensional cellular topologies. ReTAG is composed of three key components: a *Cellular Representation and Knowledge Extraction* module that lifts input graphs into cellular complexes and constructs a cell complex knowledge base; a *Retrieval-Augmented Graph Inference* module that retrieves semantically and topologically aligned subcomplexes to guide graph learning; and a *Cellular Topological Contrastive Learning* module that regularizes learning and enhances generalization through topology-aware objectives.

A. Cellular Representation and Knowledge Extraction

a) Fundamental Cycle-Guided Complex Lifting: We construct a 2-dimensional regular cell complex from the task graph $G = (V, E, T)$ by first treating it as a 1-dimensional Cell complex, where: Each vertex $v \in V$ is a 0-cell, forming the 0-skeleton $X^{(0)}$; Each edge $(u, v) \in E$ is a 1-cell attached to the 0-skeleton, yielding the 1-skeleton:

$$X^{(1)} = X^{(0)} \cup \{e \mid (u, v) \in E\}. \quad (1)$$

To identify higher-dimensional structures, we first fix a spanning tree $\mathcal{T} \subseteq G$. Since $\mathcal{T}$ is contractible, we can collapse it to a single point via the quotient map:

$$\gamma : G \rightarrow G/\mathcal{T}, \quad (2)$$

which collapses the entire subtree $\mathcal{T}$ to a single point. Here, $G/\mathcal{T}$ denotes the quotient space obtained by identifying all vertices and edges of $\mathcal{T}$ to a single point.

Under this contraction, each edge $e = (u, v) \in E \setminus \mathcal{T}$ corresponds to a fundamental cycle formed by adding e to the collapsed tree point, as the non-tree edges become loops attached to that point. The preimage $\gamma^{-1}(e)$ recovers the unique cycle c_e in the original graph, consisting of e together with the tree path in $\mathcal{T}$. For each such cycle, we attach a 2-cell by choosing an attaching map:

$$\varphi_e : \partial D^2 \cong S^1 \longrightarrow c_e \subset X^{(1)}, \quad (3)$$

and extending it via a characteristic map

$$\Phi_e : D^2 \longrightarrow X, \quad \Phi_e|_{\partial D^2} = \varphi_e,$$

where the attaching map φ_α glues the boundary of the 2-cell D^2 onto the 1-skeleton $X^{(1)}$, $x_e^2 = \Phi_e(D^2)$ defines the 2-cell corresponding to edge e .

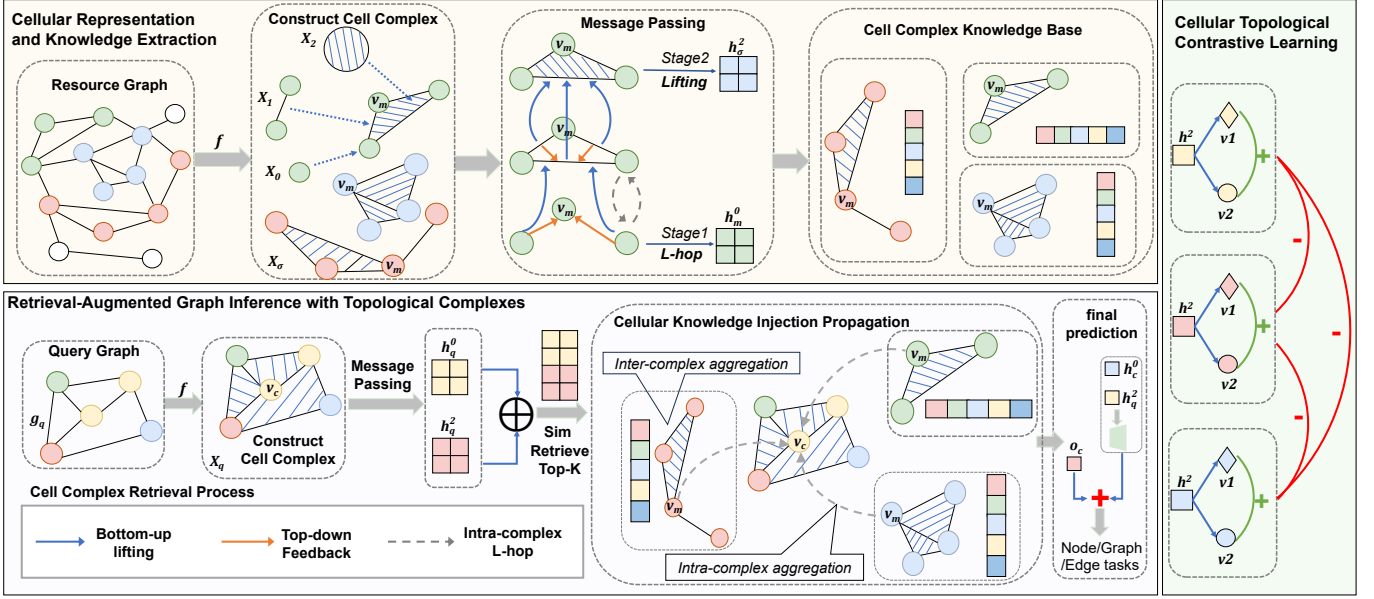


Fig. 2. The overview of Retrieval Cellular Topologies-Augmented Graph Learning Framework.

Formally, the collection of these 2-cells constitutes

$$X^{(2)} = \{x_e^2 \simeq D^2 \mid e = (u, v) \in E \setminus \mathcal{T}\}, \quad (4)$$

where each disk D^2 is attached along its boundary $\partial D^2 \cong S^1$ via the loop $c_e \in X^{(1)}$.

Proposition 1: $G/\mathcal{T}$ is homotopy-equivalent to G , and γ induces an isomorphism on the first homology group $H_1(G; \mathbb{Z})$. **Contractibility of the spanning tree.** A spanning tree $\mathcal{T}$ is connected and acyclic, hence it is contractible. In topological terms, a contractible subspace can be continuously shrunk to a point within itself.

Collapsing a contractible subspace. Consider the quotient map $\gamma : G \rightarrow G/\mathcal{T}$ that identifies all points of $\mathcal{T}$ to a single vertex v_0 . Collapsing a contractible subspace is a deformation retraction up to homotopy: there exists a continuous map $r : G \rightarrow G/\mathcal{T}$ and a homotopy $H : G \times [0, 1] \rightarrow G$ such that $H(x, 0) = x$ and $H(x, 1) = r(x)$ for all $x \in G$, with $r \circ \gamma \simeq \text{id}_{G/\mathcal{T}}$. Therefore, G and $G/\mathcal{T}$ are homotopy-equivalent.

Induced isomorphism on first homology. Homotopy-equivalent spaces have isomorphic homology groups. Hence the induced map $\gamma_* : H_1(G; \mathbb{Z}) \rightarrow H_1(G/\mathcal{T}; \mathbb{Z})$ is an isomorphism.

Intuition in graph terms. The spanning tree $\mathcal{T}$ contains no cycles, so collapsing it does not remove or merge any cycles in G . Each fundamental cycle in G (formed by a non-tree edge and the unique tree path connecting its endpoints) is preserved in $G/\mathcal{T}$ as a loop based at the collapsed tree vertex. Therefore, the first homology group $H_1(G)$ — which measures independent cycles — remains unchanged.

Proposition 2: Each non-tree edge $e \in E \setminus \mathcal{T}$ induces a unique fundamental cycle in G , which becomes a nontrivial loop in $G/\mathcal{T}$. The collection of these loops forms a basis of the first homology group $H_1(G; \mathbb{Z})$, capturing all independent cycles and providing a concise topological summary of the graph.

Let $G = (V, E)$ be a finite connected graph and $\mathcal{T} \subset G$ a spanning tree.

Existence and uniqueness of fundamental cycles. For each non-tree edge $e = (u, v) \in E \setminus \mathcal{T}$, there exists a unique simple path $P_{\mathcal{T}}(u, v)$ in $\mathcal{T}$ connecting u and v , by acyclicity of $\mathcal{T}$. Concatenating e with $P_{\mathcal{T}}(u, v)$ defines a unique simple cycle:

$$C_e = e \cup P_{\mathcal{T}}(u, v) \subset G. \quad (5)$$

Under the quotient map $\gamma : G \rightarrow G/\mathcal{T}$ that collapses $\mathcal{T}$ to a point, the tree-path $P_{\mathcal{T}}(u, v)$ is mapped to that point, so C_e becomes a nontrivial loop in $G/\mathcal{T}$.

Spanning and independence in homology. The cyclomatic number of G is $\beta_1(G) = |E| - |V| + 1$, which equals the number of non-tree edges. Thus there are exactly $\beta_1(G)$ fundamental cycles.

Any cycle in G can be expressed as a linear combination of these fundamental cycles: traversing a cycle, each time a non-tree edge e is encountered, the corresponding C_e can be used to eliminate segments along $\mathcal{T}$.

These cycles are independent in $H_1(G)$, because under γ , each fundamental cycle maps to a distinct loop in $G/\mathcal{T}$, and loops around different edges of a wedge of circles are linearly independent in homology. Hence the set $\{[C_e] \mid e \in E \setminus \mathcal{T}\}$ forms a basis of $H_1(G)$.

Topological summary. Different choices of spanning tree yield different sets of fundamental cycles as edge sets, but the corresponding homology classes always span $H_1(G)$. Therefore, these loops capture all independent cyclic dependencies of G , providing a concise topological summary suitable for lifting G into a higher-dimensional cell complex.

b) Multi-dimensional Topological Message Passing (MTMP): We design a two-stage message passing mechanism leveraging the hierarchical structure of the cell complex. The first stage propagates information along the 1-skeleton via alternating message exchanges between nodes (0-cells) and

edges (1-cells) over L hops. The second stage allows cells of all dimensions to propagate and aggregate high-dimensional information via their attached higher-dimensional cells (2-cells), enabling rich topological context exchange across different cell dimensions.

To model these interactions, we define three message passing operations over each cell complex x , corresponding to its faces, cofaces, and adjacent complexes:

$$\begin{aligned} m_{\mathcal{F}}^{l+1}(x) &= \text{AGG}_{w \in \mathcal{F}(x)} (M_{\mathcal{F}}(h_x^l, h_w^l)), \\ m_{\mathcal{C}}^{l+1}(x) &= \text{AGG}_{w \in \mathcal{C}(x)} (M_{\mathcal{C}}(h_x^l, h_w^l)), \\ m_{\uparrow}^{l+1}(x) &= \text{AGG}_{w \in \mathcal{N}_{\uparrow}(x)} (M_{\uparrow}(h_x^l, h_w^l, h_{x \cup w}^l)), \end{aligned} \quad (6)$$

where l indicates the iteration step, $\mathcal{F}(x) = \{y \mid y \prec x\}$ and $\mathcal{C}(x) = \{z \mid x \prec z\}$ denote the sets of faces and cofaces of complex x , respectively, with $\prec$ representing the face relation (i.e., $y \prec x$ means y is a face of x). The set $\mathcal{N}_{\uparrow}(x)$ contains complexes adjacent to x via a shared coface.

We incorporate the above message types into a two-stage message propagation scheme: The first stage enables a topological expansion over 1-skeleton between 0-cell and 1-cell complexes. The representation is updated as:

$$h_x^l = \text{UPDATE}(h_x^l, m_{\mathcal{F}}^l(x), m_{\mathcal{C}}^l(x)), \quad (7)$$

where the first-stage message passing iterates up to L hops.

The second stage integrates the three message types, allowing cells of all dimensions to propagate and aggregate high-dimensional information through their attached 2-cells, thereby enriching the representation as:

$$h_x^{(L+1)} = \text{UPDATE}(h_x^L, m_{\mathcal{F}}^L(x), m_{\mathcal{C}}^L(x), m_{\uparrow}^{L+1}(x)). \quad (8)$$

c) Cell Complex Knowledge Base Construction: Finally, we construct the cell complex knowledge base $X = \{X_{\sigma}\}$. To mitigate the dominance of high-degree 0-cells (i.e., nodes) and better capture long-tail topological patterns, we adopt an inverse importance sampling strategy for selecting seed 0-cell complexes. The importance score of each 0-cell complex x is defined as a convex combination of its PageRank score $\text{PR}(x)$ and its degree $\text{deg}(x)$ using the formula $I(x) = \alpha \cdot \text{PR}(x) + (1 - \alpha) \cdot \text{deg}(x)$, where $\alpha \in [0, 1]$ balances the relative weight between centrality and connectivity. Sampling probabilities are computed by normalized inverse importance:

$$p(x_i) = \frac{1/(I(x_i) + \varepsilon)}{\sum_j 1/(I(x_j) + \varepsilon)}, \quad (9)$$

where $\varepsilon > 0$ is a small constant ensuring numerical stability.

Each local cell complex $\mathcal{X}_{\sigma}$ is defined as the k -hop neighborhood around the master 0-cell x_m^0 , i.e., the 0-cell with the highest degree in the complex. It comprises all 0-cells and 1-cells reachable within k hops from x_m^0 , along with all 2-cells having at least one 1-cell face within this neighborhood. To enhance structural diversity while preserving higher-dimensional topology, we apply topology-aware augmentations—such as node dropout, edge rewiring, and Gaussian perturbation—under the constraint that no 2-cell is disrupted. These augmentations generate multiple structurally consistent instances of the local complex, facilitating robust learning of topological representations.

For each local complex X_{σ} , the key is defined as $k_b = [\tau_b, h_m^0, h_{\sigma}^2]$, where τ_b is the current time step, h_m^0 denotes the embedding of the master 0-cell, and h_{σ}^2 is a pooled two-dimensional topological feature summarizing all 2-cells within X_{σ} . The corresponding value is $v_b = \{[o_i, h_i] \mid x_i \in X_{\sigma}\}$, containing the output vectors o_i and embeddings h_i of the constituent cell complexes.

B. Retrieval-Augmented Graph Inference with Topological Complexes

a) Cell Complex Retrieval Process: Given a query graph $\mathcal{G}_q$, we first apply the lifting map $f: \mathcal{G} \rightarrow X$ and employ the Multi-dimensional Topological Message Passing framework to obtain its cell complex representations. The query key is defined as $k_q = [\tau_q, h_c^0, h_q^2]$, where τ_q denotes the time step, h_c^0 is the embedding of the central 0-cell, and h_q^2 represents a pooled two-dimensional topological feature summarizing all 2-cells within the query complex X_q . The $\text{top}K$ cell complexes $X_{\text{top}K}$ are selected based on a similarity score computed as a weighted combination of the factors:

$$S(k_q, k_b) = \mathbf{w} \times [S_{\tau}(\tau_q, \tau_b), S_0(h_c^0, h_m^0), S_2(h_q^2, h_{\sigma}^2)]^T, \quad (10)$$

where $\mathbf{w} = [\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3]$ denotes a set of hyperparameter weights, $S_{\tau}(\tau_q, \tau_b) = e^{-\eta|\tau_q - \tau_b|}$ captures temporal similarity with a decay rate η , and both $S_0(\cdot, \cdot)$ and $S_2(\cdot, \cdot)$ compute cosine similarity between the corresponding features.

b) Cellular Knowledge Injection Propagation: Within each retrieved cell complex $X_{\sigma} \in X_{\text{top}K}$, we perform *intra-complex aggregation* to propagate information from the constituent cells to the master 0-cell using the proposed Multi-dimensional Topological Message Passing (MTMP) module. Specifically, both the task-specific output vectors o_i and hidden embeddings h_i of the constituent cells $x_i \in X_{\sigma}$ are aggregated and transmitted to the master 0-cell x_m^0 :

$$\mathbf{z}_m^0 = \text{MTMP}(\{\mathbf{z}_i \mid x_i \in X_{\sigma}\}), \quad (11)$$

where $\mathbf{z}_i \in \{o_i, h_i\}$ and $\mathbf{z}_m^0 \in \{o_m^0, h_m^0\}$. For parameter-free settings, such propagation can be pre-computed and cached during knowledge base construction to improve efficiency during inference.

Then we perform *inter-complex aggregation*, where the retrieved master 0-cells x_m^0 from the selected cell complexes $X_{\sigma} \in X_{\text{top}K}$ act as knowledge anchors to enhance the query center 0-cell x_c^0 in X_q . The MTMP module is reused to inject knowledge from each x_m^0 into x_c^0 . Although the retrieval and connection occur at the 0-cell level, the injected information is further propagated to higher-dimensional cells through multi-level message passing over the entire cell complex. Formally, for each query center cell:

$$h = \text{MTMP}(\{h_i \mid x_i \in X_q \cup \{x_m^0\}\}), \quad (12)$$

$$o_c = \text{MTMP}(\{o_i^0 \mid x_i^0 \in \{x_m^0\}\}), \quad (13)$$

where $h = [\{h^0\}, \{h^1\}, \{h^2\}]$ concludes embedding of various dimensional cell complexes.

Finally, at the data fusion layer, the aggregated hidden state h_c^0 of the center 0-cell x_c^0 and h_q^2 the pooling of 2

cell complexes is passed through a multi-layer perceptron $\text{MLP}(\cdot)$, yielding a structure-aware output. This output is then combined with the task-specific output o_c , which has already integrated multi-dimensional signals through the retrieval-enhanced process. The final prediction is computed as:

$$\hat{o}_c = \gamma o_c + (1 - \gamma) \text{MLP}([h_c^0, h_q^2]), \quad (14)$$

where $\gamma \in [0, 1]$ is a reweighting hyperparameter. The resulting representation $\hat{o}_c$ serves as the input to downstream node-, edge-, or graph-level tasks, typically evaluated via a task-specific similarity function.

C. Cellular Topological Contrastive Learning (CTCL)

We propose **Cellular Topological Contrastive Learning (CTCL)** to encode high-dimensional structural semantics within each 2-cell $x_k^2 \in X^{(2)}$ by enforcing feature consistency among its constituent 0-cells. Formally, let $\mathcal{X}_k^{(0)}$ denote the set of 0-cells forming x_k^2 . For each 2-cell, we independently sample two subsets of 0-cells, $\tilde{\mathcal{X}}_{k,1}^{(0)}$ and $\tilde{\mathcal{X}}_{k,2}^{(0)}$, using degree-based importance sampling. The embeddings of these subsets are aggregated to produce two representations per 2-cell:

$$\tilde{h}_{k,v} = \frac{1}{|\tilde{\mathcal{X}}_{k,v}^{(0)}|} \sum_{x^0 \in \tilde{\mathcal{X}}_{k,v}^{(0)}} (h_{x^0} + \alpha \cdot \mathcal{N}(0, 1)), \quad (15)$$

$$k = 1, \dots, N, \quad v \in \{1, 2\},$$

where h_{x^0} is the embedding of 0-cell x^0 , α controls the magnitude of Gaussian perturbation, and $N = |X^{(2)}|$ is the number of 2-cells. We adopt an InfoNCE loss to enforce consistency between these two views:

$$\mathcal{L}_{\text{CTCL}} = -\frac{1}{N} \sum_{k=1}^N \log \frac{\exp(\text{sim}(\tilde{h}_{k,1}, \tilde{h}_{k,2})/\tau)}{\sum_{j \neq k} \exp(\text{sim}(\tilde{h}_{k,1}, \tilde{h}_{j,2})/\tau)}, \quad (16)$$

where $\text{sim}(\cdot, \cdot)$ denotes the cosine similarity and τ is a temperature parameter. The overall training objective combines standard classification with topological contrastive regularization:

$$\mathcal{L}_{\text{cls}} = -\sum_{c=1}^C y_c \log \hat{y}_c, \quad (17)$$

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{cls}} + \lambda \cdot \mathcal{L}_{\text{CTCL}}, \quad (18)$$

where y_c and $\hat{y}_c$ denote the ground-truth label and predicted probability for class c , respectively, and λ balances supervised learning and topological contrastive regularization.

V. EXPERIMENTS

A. Experimental Setup

a) Datasets: We evaluate ReTAG on both static graphs and dynamic interaction graphs. The static datasets cover molecular and protein graphs with varying sizes and label settings, while the dynamic datasets consist of large-scale temporal interaction networks used for link prediction.

Static Datasets. We consider four widely used benchmarks for graph-level and node-level prediction tasks:

TABLE I
STATISTICS OF THE EXPERIMENTAL DATASETS.

Statistics	Dynamic Graphs		Static Graphs			
	KOUBEI	AMAZON	PROTEINS	COX2	ENZYMES	BZR
Nodes / Graph	221,366	238,735	39.06	41.22	32.63	35.75
Edges / Graph	3986609	876237	72.82	43.45	62.14	38.36
Density	3.3e-4	6.2e-5	4.8e-2	2.6e-2	5.9e-2	3.0e-2
Graphs	1	1	1,113	467	600	405
Graph Classes	—	—	2	2	6	2
Node Features	—	—	3	1	18	3
Task	Edge	Edge	Node, Graph	Graph	Node, Graph	Graph

- **PROTEINS** [50]: A dataset of 1,113 protein graphs representing structural interaction patterns. It is used for both node classification and graph classification tasks.
- **COX2** [51]: A molecular graph dataset containing 467 chemical compounds related to COX-2 inhibitors. It is used for binary graph classification.
- **ENZYMES** [52]: A benchmark of 600 enzyme graphs categorized into six functional classes. It supports both node-level and graph-level classification.
- **BZR** [51]: A dataset of 405 molecular graphs describing ligands for the benzodiazepine receptor. It is used for binary graph classification.

Dynamic Datasets. For temporal link prediction, we employ two large-scale dynamic interaction graphs, constructed from time-ordered interaction logs:

- **KOUBEI**: A nine-week user–store interaction graph containing hundreds of thousands of nodes and several million edges. It is used for temporal edge prediction.
- **AMAZON**: A product–user interaction graph derived from weekly Amazon review logs. It contains more than two hundred thousand nodes and is used for dynamic link prediction.

A summary of the dataset statistics, including graph sizes, feature dimensions, and task types, is provided in Table I.

b) Comparison Methods: We compare ReTAG with representative baselines for both dynamic and static graph learning tasks.

Dynamic Graphs.

- *GraphCL-based Methods:* **LightGCN** [53] linearly propagates embeddings for user–item recommendation. **SGL** [54] adds self-supervised discrimination to enhance representation learning. **MixGCF** [55] mixes synthetic negatives across layers for collaborative filtering. **SimGCL** [56] applies contrastive learning for robust dynamic link prediction. **StabCF** [57] proposes a stabilized training strategy for collaborative filtering that mitigates gradient oscillation in GNN-based recommenders.
- *Prompt-based Methods:* **GraphPro** [58] introduces spatial–temporal prompts for dynamic graphs. **GraphPro+PRODIGY** [59] enhances prompting with in-context learning.
- *Retrieval-Augmented Methods:* **RAGraph** [13] retrieves similar graphs via multi-dimensional similarity and injects external structural knowledge into the query graph.

Static Graphs.

- *GNN Baselines*: **GCN** [60] aggregates neighborhood information for node/graph classification. **GraphSAGE** [61] generates inductive embeddings via neighbor sampling. **GAT** [62] uses attention-weighted neighbor aggregation. **GIN** [63] applies MLP-based injective aggregation for expressive graph representations.
- *Retrieval-Augmented Variant*: **RAGraph** [13] integrates retrieved structural priors to enhance GNN learning.
- *Prompt-based Variants*: **GraphPrompt** [64] employs task-specific prompts for unified graph pre-training. **GraphPrompt+PRODIGY** [59] introduces contextual prompting with in-context examples. **ProNoG** [65] generates node-specific prompts via multi-hop non-homophily patterns. **GP2F** [66] performs cross-domain graph prompting with adaptive fusion of multiple pre-trained GNNs for few-shot graph adaptation.

c) *Evaluation Metrics*: For node and graph classification tasks, we report the classification accuracy. For dynamic link prediction, we follow prior work and use top- K ranking metrics, including Recall@20 and nDCG@20, computed on the held-out test snapshots.

d) *Parameters Settings*: We adopt a training–resource split: model training is performed on the labeled data, while retrieval is conducted from a disjoint resource set. For static graphs, we use a 50%/30%/20% split of nodes (train/validation/test). For dynamic graphs, interaction snapshots are chronologically partitioned, and only past snapshots are used as the resource set for time-aware retrieval. Retrieval-based methods (e.g., PRODIGY, RAGraph) retrieve exclusively from the resource set to prevent information leakage, whereas other baselines are fine-tuned on the merged graph. All experiments are implemented in PyTorch and run on a single NVIDIA A100 GPU (40GB). We use the Adam optimizer with learning rate 1×10^{-3} and batch size 16. For classification tasks, the hidden dimension is set to 256, and the first stage of MTMP employs $L = 2$ layers. Cell complexes are constructed from k -hop neighborhoods, where k is selected on the validation set (typically $k \in \{2, 3\}$). The CTCL module applies a 2-dimensional contrastive loss with weight $\lambda \in [0.1, 0.4]$. For link prediction, the first stage of MTMP adopts a 3-layer architecture ($L = 3$) with 64-dimensional embeddings, and the whole model is trained end-to-end with both contrastive and retrieval modules.

B. Cellular Complex-Augmented Graph Results

The experimental results of our model and baselines on node and graph classification for static graphs and on link prediction for dynamic graphs are summarized in Table II and Table III. Based on the comparison with existing baselines, we draw the following observations:

- **Our proposed ReTAG achieves the best performance.** ReTAG significantly outperforms all baselines on both static and dynamic graphs. On static datasets, ReTAG achieves a relative improvement of 2.3%–3.7% in node classification accuracy and 1.9%–4.4% in graph classification accuracy over state-of-the-art methods, consistently achieving the highest performance across all

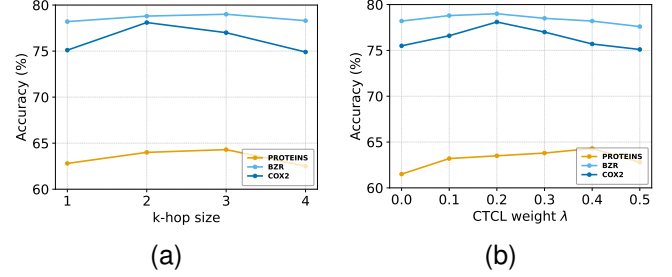


Fig. 3. Accuracy under different settings. (a) Varying k -hop size. (b) Varying CTCL weight.

tasks. On dynamic benchmarks, ReTAG achieves relative gains of 2.7%–6.7% in Recall@20 and 2.3%–6.2% in nDCG@20 over strong baselines such as RAGraph/FT. These gains can be attributed to ReTAG’s ability to incorporate high-dimensional topological knowledge via cell complex lifting and retrieve structured support from external graphs. Its MTMP module enables hierarchical propagation across complex structures, while the CTCL loss regularizes representations by enforcing 2-cell consistency, jointly enhancing expressiveness and generalizability across tasks.

- **Retrieval-augmented topology reasoning is crucial for generalized graph learning.** From Table II and Table III, we observe that retrieval-augmented baselines (e.g., RAGraph/FT) consistently outperform standard GNNs and prompt-based variants, demonstrating that incorporating external knowledge via retrieval is vital for generalization. However, these baselines often rely on shallow node- or edge-level similarity, limiting their ability to capture complex structural semantics. ReTAG addresses this limitation by lifting graphs into higher-dimensional cell complexes and conducting retrieval based on both semantic and topological alignment. This enables the model to obtain more informative and transferable support subgraphs. With MTMP injection and CTCL regularization, ReTAG better integrates retrieved knowledge, yielding superior performance on various graph tasks.

C. Hyper-parameter Sensitivity Study

We investigate the sensitivity of ReTAG to two key hyper-parameters: the k -hop neighborhood size used for constructing cell complexes, and the cellular topological contrastive learning (CTCL) loss weight. These hyper-parameters respectively determine the receptive field of the lifted complex and the strength of 2-cell topological regularization. We vary k from 1 to 4 and the CTCL loss weight from 0 to 0.5 to analyze their individual effects. The results are presented in Fig. 3.

- **k -hop Neighborhood Size.** Fig. 3(a) shows the impact of varying the k -hop receptive field. As k increases, each cell complex incorporates richer structural context and enables retrieval of larger supporting subgraphs. However, excessively large k introduces noisy or redundant structures that degrade performance, while overly small k fails to capture essential topological motifs for retrieval

TABLE II

ACCURACY EVALUATION ON NODE AND GRAPH CLASSIFICATION. ALL TABULAR RESULTS (%) ARE REPORTED AS MEAN $\pm$ STD ACROSS FIVE RUNS. THE BEST RESULTS ARE **BOLDED** AND THE RUNNER-UPS ARE UNDERLINED.

Methods	Node Classification		Graph Classification			
	PROTEINS (5-shot)	ENZYMES (5-shot)	PROTEINS (5-shot)	COX2 (5-shot)	BZR (5-shot)	ENZYMES (5-shot)
<i>GNN Baselines</i>						
GCN	46.63 $\pm$ 03.04	52.80 $\pm$ 12.89	54.80 $\pm$ 06.64	67.87 $\pm$ 03.39	58.76 $\pm$ 05.08	22.67 $\pm$ 05.20
GraphSAGE	48.87 $\pm$ 02.64	48.75 $\pm$ 01.59	54.68 $\pm$ 09.34	67.02 $\pm$ 05.42	58.27 $\pm$ 04.79	21.17 $\pm$ 05.49
GAT	48.13 $\pm$ 07.90	47.75 $\pm$ 01.23	55.82 $\pm$ 07.31	64.89 $\pm$ 03.23	57.04 $\pm$ 06.70	20.67 $\pm$ 03.27
GIN	49.61 $\pm$ 01.58	48.82 $\pm$ 01.58	56.17 $\pm$ 08.58	62.77 $\pm$ 02.85	56.54 $\pm$ 04.20	21.10 $\pm$ 02.53
<i>GraphPrompt+ Variants</i>						
Vanilla/NF	44.88 $\pm$ 13.17	48.81 $\pm$ 01.88	56.68 $\pm$ 03.63	53.04 $\pm$ 04.13	68.77 $\pm$ 03.44	36.50 $\pm$ 03.31
Vanilla/FT	48.99 $\pm$ 01.88	51.99 $\pm$ 01.36	57.04 $\pm$ 03.88	64.04 $\pm$ 08.20	69.01 $\pm$ 02.21	40.00 $\pm$ 04.36
PRODIGY/NF	47.32 $\pm$ 08.12	43.80 $\pm$ 14.03	53.48 $\pm$ 06.72	53.97 $\pm$ 10.34	67.18 $\pm$ 08.93	22.12 $\pm$ 13.84
PRODIGY/FT	53.26 $\pm$ 06.42	57.98 $\pm$ 12.37	57.14 $\pm$ 10.34	65.31 $\pm$ 04.28	68.08 $\pm$ 06.68	25.94 $\pm$ 05.12
ProNoG	52.89 $\pm$ 11.76	<u>76.48</u> $\pm$ 06.23	61.63 $\pm$ 08.01	58.03 $\pm$ 14.28	62.26 $\pm$ 12.27	37.91 $\pm$ 05.64
GP2F	58.43 $\pm$ 02.07	49.69 $\pm$ 01.35	58.88 $\pm$ 01.39	56.73 $\pm$ 07.68	58.75 $\pm$ 10.93	23.25 $\pm$ 02.78
<i>Retrieval-Augmented Variants</i>						
RAGRAPHS/NF	56.12 $\pm$ 04.11	75.92 $\pm$ 01.72	58.48 $\pm$ 03.93	55.32 $\pm$ 04.15	77.53 $\pm$ 05.26	38.17 $\pm$ 03.39
RAGRAPHS/FT	<u>58.74</u> $\pm$ 00.87	75.74 $\pm$ 01.92	<u>62.33</u> $\pm$ 02.52	<u>76.60</u> $\pm$ 02.30	76.79 $\pm$ 05.02	<u>47.71</u> $\pm$ 06.88
ReTAG(Ours)	60.91 $\pm$ 01.79	78.26 $\pm$ 01.57	64.30 $\pm$ 02.21	78.09 $\pm$ 02.57	79.01 $\pm$ 03.49	49.83 $\pm$ 04.72

TABLE III
PERFORMANCE EVALUATION (%) ON LINK PREDICTION.

Methods	KOUBEI		AMAZON	
	Recall@20	nDCG@20	Recall@20	nDCG@20
<i>GraphCL</i>				
LightGCN	30.21 $\pm$ 06.45	22.24 $\pm$ 05.83	15.07 $\pm$ 06.48	06.53 $\pm$ 02.66
SGL	32.61 $\pm$ 04.27	22.36 $\pm$ 04.82	15.78 $\pm$ 07.12	07.90 $\pm$ 02.49
MixGCF	32.06 $\pm$ 04.20	22.49 $\pm$ 06.91	15.24 $\pm$ 08.98	07.40 $\pm$ 03.44
SimGCL	33.07 $\pm$ 05.28	23.08 $\pm$ 05.55	16.10 $\pm$ 07.91	07.58 $\pm$ 03.51
StableCF	32.42 $\pm$ 02.01	17.55 $\pm$ 01.37	13.59 $\pm$ 05.28	05.27 $\pm$ 04.64
<i>GraphPro+</i>				
Vanilla/NF	21.31 $\pm$ 04.59	15.31 $\pm$ 03.11	12.56 $\pm$ 07.45	06.31 $\pm$ 03.92
Vanilla/FT	33.96 $\pm$ 04.13	24.66 $\pm$ 02.78	18.14 $\pm$ 07.55	08.73 $\pm$ 03.74
PRODIGY/NF	21.66 $\pm$ 03.21	14.82 $\pm$ 03.92	11.88 $\pm$ 02.61	05.84 $\pm$ 01.84
PRODIGY/FT	33.46 $\pm$ 04.70	23.28 $\pm$ 03.40	16.72 $\pm$ 04.28	08.09 $\pm$ 02.66
<i>Retrieval-Augmented</i>				
RAGRAPHS/NF	22.86 $\pm$ 03.44	16.68 $\pm$ 02.48	13.78 $\pm$ 05.54	06.52 $\pm$ 02.69
RAGRAPHS/FT	<u>34.27</u> $\pm$ 03.93	<u>24.82</u> $\pm$ 02.69	<u>18.32</u> $\pm$ 07.45	<u>09.09</u> $\pm$ 03.89
ReTAG (Ours)	35.21 $\pm$ 03.21	25.39 $\pm$ 02.23	19.54 $\pm$ 08.04	09.65 $\pm$ 04.06

and reasoning. This highlights the importance of selecting a moderate k to balance completeness and noise.

- **CTCL Loss Weight λ .** Fig. 3(b) examines the effect of the CTCL loss weight. When λ is too small, the model underutilizes discriminative 2-cell information, weakening topological regularization. Conversely, large values of λ overly constrain the optimization and conflict with the primary learning objective, resulting in reduced accuracy. A moderate CTCL weight therefore provides the best trade-off between topological consistency and task-specific learning.

D. Few-shot Generalization Study

We evaluate the few-shot performance of ReTAG under 1–5 shot settings across four datasets (PROTEINS, BZR,

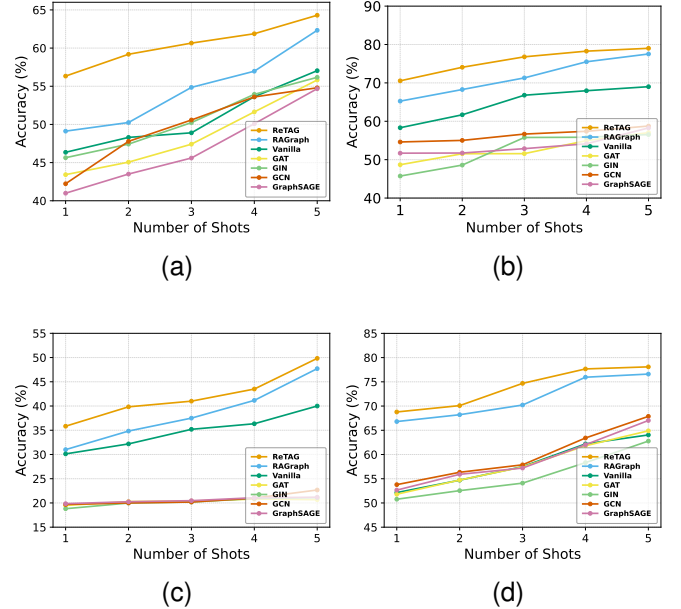


Fig. 4. Few-shot accuracy comparison of ReTAG and baseline methods (RAGraph, Vanilla, GAT, GIN, GCN, GraphSAGE) across four datasets: (a) PROTEINS, (b) BZR, (c) ENZYMES, and (d) COX2. Results are reported under the 1–5 shot setting.

ENZYMES, COX2), together with representative baseline methods (RAGraph, Vanilla, GAT, GIN, GCN, GraphSAGE). As shown in Fig. 4, accuracy generally improves as more labeled samples become available, following the typical trend of metric-based adaptation under low-data regimes. Across all datasets and shot levels, ReTAG maintains consistently strong performance and remains competitive as supervision increases. These results indicate that the proposed retrieval-augmented topological modeling is effective in scenarios with limited labeled data.

TABLE IV
ABLATION STUDY ON COX2 AND PROTEINS DATASETS.

Model Variant	COX2 (%)	PROTEINS (%)
w/o FCL	76.05±02.89	62.33±02.52
w/o MTMP	76.21±01.97	62.45±03.19
w/o CTCL	76.94±01.83	63.21±04.79
Full Model (ReTAG)	78.09±02.57	64.30±02.21

E. Ablation Study

We evaluate three ablated variants of ReTAG: removing the Fundamental Cycle-Guided Complex Lifting (FCL), which eliminates 2-cell construction and reduces the model to an edge-based graph; discarding the Multi-dimensional Topological Message Passing (MTMP), which disables hierarchical propagation across cell dimensions; and removing the Cellular Topological Contrastive Learning (CTCL), which drops contrastive regularization over 2-cell representations. As shown in Table IV, we observe that:

- The performance drops when removing FCL, as eliminating 2-cells via fundamental cycles reduces the model to a purely edge-based graph. This shows that high-dimensional cycles are crucial for capturing richer structural context and stronger structural reasoning.
- The model’s performance suffers when MTMP is discarded. Disabling MTMP restricts interactions to nodes and edges, preventing the model from reasoning over cycles and their associated higher-dimensional cells. This shows that MTMP is crucial for capturing complex topological dependencies and multi-dimensional context.
- The performance of the model drops when removing CTCL. Without contrastive regularization over 2-cell representations, the model fails to enforce consistency among high-dimensional structures, leading to less discriminative embeddings. This shows that CTCL is essential for improving robustness and generalization in capturing cellular topology.

F. Cycle Structure and Representation Analysis

a) Cycle Structure Analysis: To examine when ReTAG is expected to help, we analyze the cycle statistics of all datasets. Fig. 5(a) shows the fraction of nodes that participate in at least one cycle, revealing large variations in cycle density: molecular and protein graphs are cycle rich, whereas some benchmarks are almost tree-like, and large recommendation graphs lie in between with moderate cycle coverage. Fig. 5(b) further shows that the number of fundamental cycles spans several orders of magnitude across datasets.

Despite this diversity, ReTAG consistently outperforms baselines (Table II and Table III), demonstrating its robustness across varying cycle densities. When cycles are abundant, ReTAG lifts informative motifs to 2-cells and propagates them through MTMP; when cycles are sparse, the model naturally reduces to an edge-centric encoder while still retaining the advantages of topology-aware retrieval. Together with the ablation results, these observations show that explicitly model-

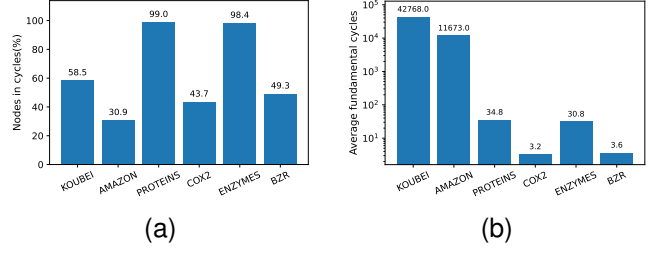


Fig. 5. Cycle statistics across datasets. (a) Fraction of nodes that belong to at least one cycle. (b) Average number of fundamental cycles per graph (log scale).

ing fundamental cycles improves representation quality across heterogeneous topological distributions.

b) Representation Visualization: To further inspect how cycle-aware lifting and MTMP shape the learned representations, we visualize graph-level embeddings using t-SNE from both real-world and controlled benchmarks. Fig. 6 first reports the visualization on ENZYMES. The standard GCN baseline in Fig. 6(a) produces a largely isotropic cloud, where graphs from different classes are highly mixed and no clear global organization is formed. This pattern suggests that the encoder mainly relies on local neighborhood statistics and has limited ability to organize graphs by their higher-order topological structures.

In contrast, the ReTAG embeddings in Fig. 6(b) concentrate along several smooth manifolds with more clearly separated branches. This indicates that ReTAG arranges enzyme graphs according to shared cycle-aware motifs: graphs with similar configurations of fundamental cycles are mapped to nearby positions along the same curve, whereas graphs with distinct cycle structures tend to occupy different segments of the manifold. Such geometry naturally arises from lifting fundamental cycles to 2-cells and propagating information jointly over nodes, edges, and 2-cells via MTMP. Although class labels are not perfectly separated in two dimensions due to the structural heterogeneity within each enzyme class, ReTAG still significantly outperforms GCN in Table II, confirming that explicit higher-dimensional cycle modeling yields more discriminative and topology-consistent graph representations.

We further validate this observation with the controlled t-SNE study in Fig. 7, generated by the motivational experiment in which two 3-regular graph classes share the same number of nodes, edges, and degree statistics but differ in cycle topology: $K_{3,3}$ contains no triangles, whereas the triangular prism contains two triangular 2-cell motifs. Under conventional message passing, GCN and GAT produce heavily overlapped embeddings and achieve only random-level accuracy ($50.07\% \pm 3.41\%$ and $49.83\% \pm 2.65\%$, respectively), while GraphSAGE shows only partial separation ($84.90\% \pm 5.21\%$). Once explicit cycle features are injected as a ReTAG-like proxy for 2-cell information, the two classes become cleanly separated in the embedding space and the accuracy reaches $100.00\% \pm 0.00\%$. This controlled visualization complements the ENZYMES result by isolating the source of improvement: cellular cycle-level information provides discriminative signals that are largely orthogonal to node/edge-level neighborhood

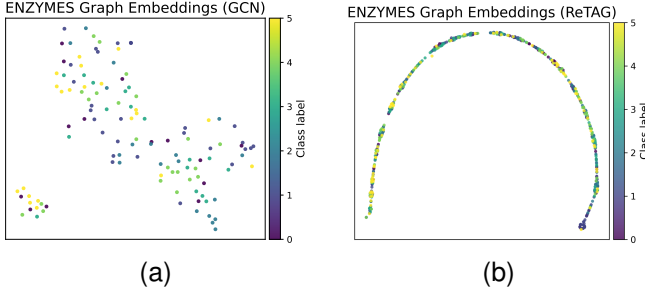


Fig. 6. t-SNE visualization of graph-level embeddings on ENZYMES. (a) The GCN baseline. (b) ReTAG.

aggregation.

G. Case Study: Topology-Aware Retrieval on COX2

To better illustrate how ReTAG exploits cellular topologies during retrieval, we conduct a case study on a representative COX2 molecular graph. The left panel of Fig. 8 shows the query graph, where three fundamental cycles (1–3) detected by FCL are lifted into 2-cells that serve as the foundational high-dimensional topological primitives for retrieval. In both the query and retrieved graphs, the yellow node marks the *master 0-cell*, whose pivotal high-degree position serves as a structural anchor to align local neighborhoods across the lifted complexes.

For the retrieved supports (right), we highlight matching 2-cells (dark-blue) and their branching environments (light-blue nodes, dashed edges). Although external to 2-cells, these branches contextualize motifs, distinguishing cycles with similar shapes but different surroundings. By explicitly visualizing this context, we verify the consistency of topological alignment around the master anchor.

As shown in the right panel, the top-1 support closely replicates all three query 2-cells with highly consistent branching patterns, whereas lower-ranked supports capture only subsets of these motifs with increasingly divergent local contexts. This case study demonstrates that ReTAG aligns graphs by jointly leveraging high-dimensional cellular structures and their surrounding branching geometry, enabling retrieval to capture multi-level topological alignment beyond local adjacency.

H. Complexity and Scalability Analysis

a) Computational Complexity: We analyze the asymptotic time and space complexity of ReTAG. Let $n = |V|$ and $m = |E|$ denote the numbers of nodes and edges in a graph, r the number of 2-cells (fundamental cycles) in the lifted complex, d the hidden dimension, L the number of message-passing layers, N_{res} the size of the resource pool, and K the number of retrieved supports.

The backbone encoder of ReTAG follows standard message passing with a per-layer cost of $\mathcal{O}(md)$. Consequently, an L -layer encoder requires $\mathcal{O}(Lmd)$ per forward pass, which is consistent with standard GCN and RAGraph-style baselines. Cell-complex lifting (spanning tree extraction and fundamental cycle basis construction) is performed offline in $\mathcal{O}(m + r)$ time and does not participate in backpropagation. During

training and inference, the MTMP module operates on the lifted complex with a complexity of $\mathcal{O}((m + r)d)$ per layer. Since complexes are constructed based on local k -hop neighborhoods, r is effectively bounded by a small constant. Thus, the overall MTMP cost remains linear with respect to m . Topology-aware retrieval computes similarities between a d -dimensional query and all N_{res} resource keys to select top- K supports, costing $\mathcal{O}(N_{\text{res}}d)$, while the cellular topological contrastive loss is $\mathcal{O}(rd)$ per batch. Therefore, the online time complexity of ReTAG is linear in terms of the graph size and N_{res} , incurring only a moderate constant-factor overhead compared to conventional GNN baselines.

b) Space Complexity: ReTAG stores node, edge, and 2-cell embeddings alongside the resource pool, resulting in a memory usage of $\mathcal{O}((n + m + r + N_{\text{res}})d)$. Thus, both the time and space complexity of ReTAG scale linearly with the graph size and the number of retrieved resources.

VI. CONCLUSION

In this work, we explore the role of higher-dimensional topological structures in retrieval-augmented graph learning. We propose a novel framework, namely Retrieved Cellular Topologies-Augmented Graph Learning (ReTAG), which lifts input graphs into cellular complexes and constructs a knowledge base of multi-dimensional topology-aware subgraphs, termed cellular topologies. During inference, ReTAG retrieves structurally aligned cellular topologies and incorporates them via a multi-dimensional message-passing mechanism that captures complex topological dependencies beyond conventional pairwise relations. Furthermore, a cellular topological contrastive learning module is introduced to reinforce high-dimensional structural semantics by aligning features within individual topological cells. Extensive experiments across multiple graph tasks demonstrate that ReTAG outperforms state-of-the-art methods.

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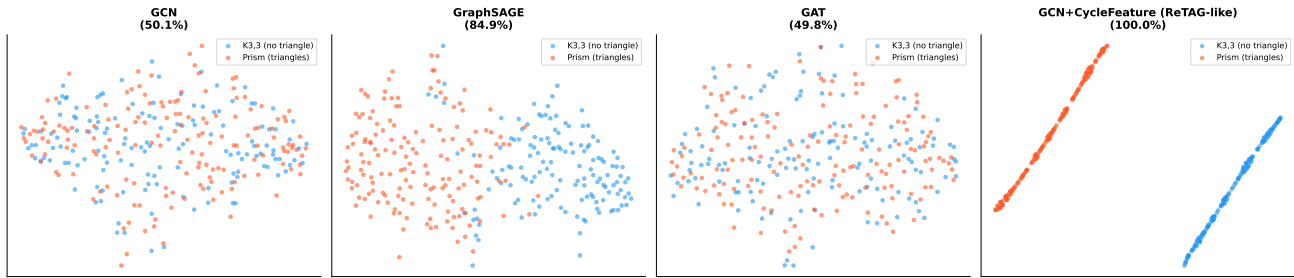


Fig. 7. Controlled t-SNE visualization on the motivational benchmark. The two classes are 3-regular graphs with identical local degree statistics but different cycle structures: $K_{3,3}$ has no triangles, whereas the triangular prism contains triangular cycles. Conventional message-passing models produce overlapped or only partially separated embeddings, while the cycle-augmented ReTAG-like variant forms clearly separated clusters.

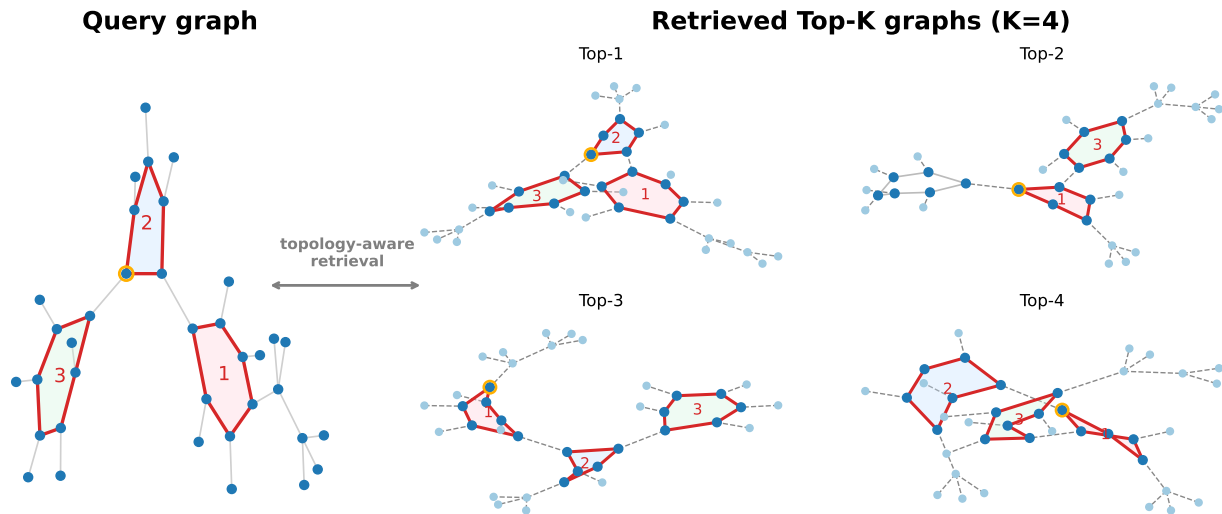


Fig. 8. Topology-aware retrieval on a molecular graph from COX2. Left: query graph with three fundamental cycles (1–3) extracted by FCL and lifted to 2-cells (shaded). Right: top-4 retrieved graphs, where shaded regions mark matching 2-cells; dark-blue nodes denote their 1-skeleton, and light-blue nodes with dashed edges show surrounding branching context. The yellow nodes mark the aligned master 0-cells across both the query and retrieved graphs.

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